

Phase 3 – Exploitation

Objective:

The objective of the exploitation phase is to verify whether the vulnerabilities identified during the scanning phase can be successfully exploited. This phase demonstrates the potential impact of the discovered vulnerabilities by attempting to gain authorized access to the target system in a controlled laboratory environment.

Purpose:

Exploitation is the third phase of penetration testing. After identifying vulnerable services and software versions during the scanning phase, the penetration tester attempts to exploit these weaknesses using appropriate tools and techniques. The primary goal is to determine whether the identified vulnerabilities can be used to gain unauthorized access or execute commands on the target system. All exploitation activities are performed only within an authorized testing environment.

Step 1: Analyze the Scanning Results:

Before attempting any exploitation, the results obtained from the scanning phase were carefully analyzed to identify services that might contain known vulnerabilities.

Activities Performed:

- Reviewed the list of open ports.
- Examined the running network services.
- Identified software versions.
- Compared the detected versions with publicly known vulnerabilities.

Observation:

The scanning results revealed several outdated services that are intentionally vulnerable in the Metasploitable2 virtual machine. These services were selected for further security assessment during the exploitation phase.

Step 2: Identify a Suitable Vulnerability:

The identified services were evaluated to determine whether publicly documented vulnerabilities were available. Public vulnerability databases and exploit frameworks can

be used by penetration testers to verify whether a service contains a known security weakness.

Activities Performed

- Identified outdated software versions.
- Matched the detected software with known vulnerabilities.
- Selected an appropriate vulnerability for testing.

Observation

A known vulnerability was selected for further testing based on the software version identified during the scanning phase.

Step 3: Plan the Exploitation Process:

Before executing any exploit, the penetration tester verifies that the target system belongs to the authorized testing environment and confirms that the selected exploitation method is appropriate for the identified vulnerability.

Activities Performed:

- Verified the target IP address.
- Confirmed the vulnerable service.
- Selected an appropriate exploitation tool.
- Planned the exploitation procedure.

Observation:

The target system and selected vulnerability were verified successfully, allowing the assessment to proceed to the practical exploitation stage.

Exploitation Summary:

The exploitation phase focuses on validating the vulnerabilities identified during the scanning phase. During this phase, the identified services are analyzed, suitable vulnerabilities are selected, and the exploitation procedure is planned before executing any practical attack. Careful planning helps ensure that testing is conducted safely, accurately, and within the scope of the authorized laboratory environment.